Arrays the Utility class

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Arrays class is an utility class to define several utility methods for Arrays or array objects.

1. Sorting elements of Array: Arrays class defines the following sort methods to sort elements of primitive type and object type:
   a. public static void sort(primitive_data_type []p): to sort the array in default natural sorting order.

   b. public static void sort(Object []p): to sort the array in default natural sorting order.

   c. public static void sort(Object []p, Comparator c): to sort the array in customized sorting order.

```
package Chapter8;
import java.util.Arrays;
import java.util.Random;
public class ArraySortDemo {
  public static void main(String[] args) {
    Random r = new Random();
    int A[] = new int[r.nextInt(10, 20)];
    for (int i = 0; i < A.length; i++) {
      A[i] = r.nextInt(10, 100);
    }
    System.out.println("Array before sorting: " + Arrays.toString(A));
    Arrays.sort(A);
    System.out.println("Array after sorting: " + Arrays.toString(A));
    Character S[] = new Character[r.nextInt(10,20)];
    String bigAlpha = "ABCDEFGHIJKLMNOPQRSTUVWXYZ";
    for (int i = 0; i < S.length; i++) {
      S[i] = bigAlpha.charAt(r.nextInt(26));
    }
    System.out.println("Character array before sorting: " + Arrays.toString(S));
    Arrays.sort(S);
    System.out.println("Character array after sorting: " + Arrays.toString(S));
    //Customized sorting order
    System.out.println("Customized sorting order starts here.");
  }
}
```

Output:

```
Array before sorting: [58, 27, 11, 91, 52, 39, 42, 70, 59, 57]
Array after sorting: [11, 27, 39, 42, 52, 57, 58, 59, 70, 91]
Character array before sorting: [I, L, Z, E, B, O, A, T, F, G, Z, D, X, V, I, N]
Character array after sorting: [A, B, D, E, F, G, I, I, L, N, O, T, V, X, Z, Z]
```

Code with Comparator Object:

```
package Chapter8;
import java.util.Arrays;
import java.util.Random;
public class CSOUsingArrays {
    public static void main(String[] args) {
        Random r = new Random();
        MyComparator c = new MyComparator();
        int A[] = new int[r.nextInt(10, 20)];
        for (int i = 0; i < A.length; i++) {
            A[i] = r.nextInt(10, 100);
        }
        System.out.println("Array before sorting: " + Arrays.toString(A));
        Arrays.sort(A); // for primitive type Comparator is not acceptable
        System.out.println("Array after sorting: " + Arrays.toString(A));
        // Customized sorting order
        System.out.println("Customized sorting order starts here.");
        Character S[] = new Character[r.nextInt(10, 20)];
        String bigAlpha = "ABCDEFGHIJKLMNOPQRSTUVWXYZ";
        for (int i = 0; i < S.length; i++) {
            S[i] = bigAlpha.charAt(r.nextInt(26));
        }
        System.out.println("Character array before sorting: " + Arrays.toString(S));
        Arrays.sort(S, c);
        System.out.println("Character array after sorting in CSO: " + Arrays.toString(S));
    }
}
```

Output:

```
Array before sorting: [83, 22, 49, 94, 71, 85, 21, 92, 31, 61, 84, 36, 91, 80]
Array after sorting: [21, 22, 31, 36, 49, 61, 71, 80, 83, 84, 85, 91, 92, 94]
Customized sorting order starts here.
Character array before sorting: [I, T, B, X, W, W, Q, Y, H, W, F, C, N, F, J, O, J, N, H]
Character array after sorting in CSO: [Y, X, W, W, W, T, Q, O, N, N, J, J, I, H, H, F, F, C, B]
```